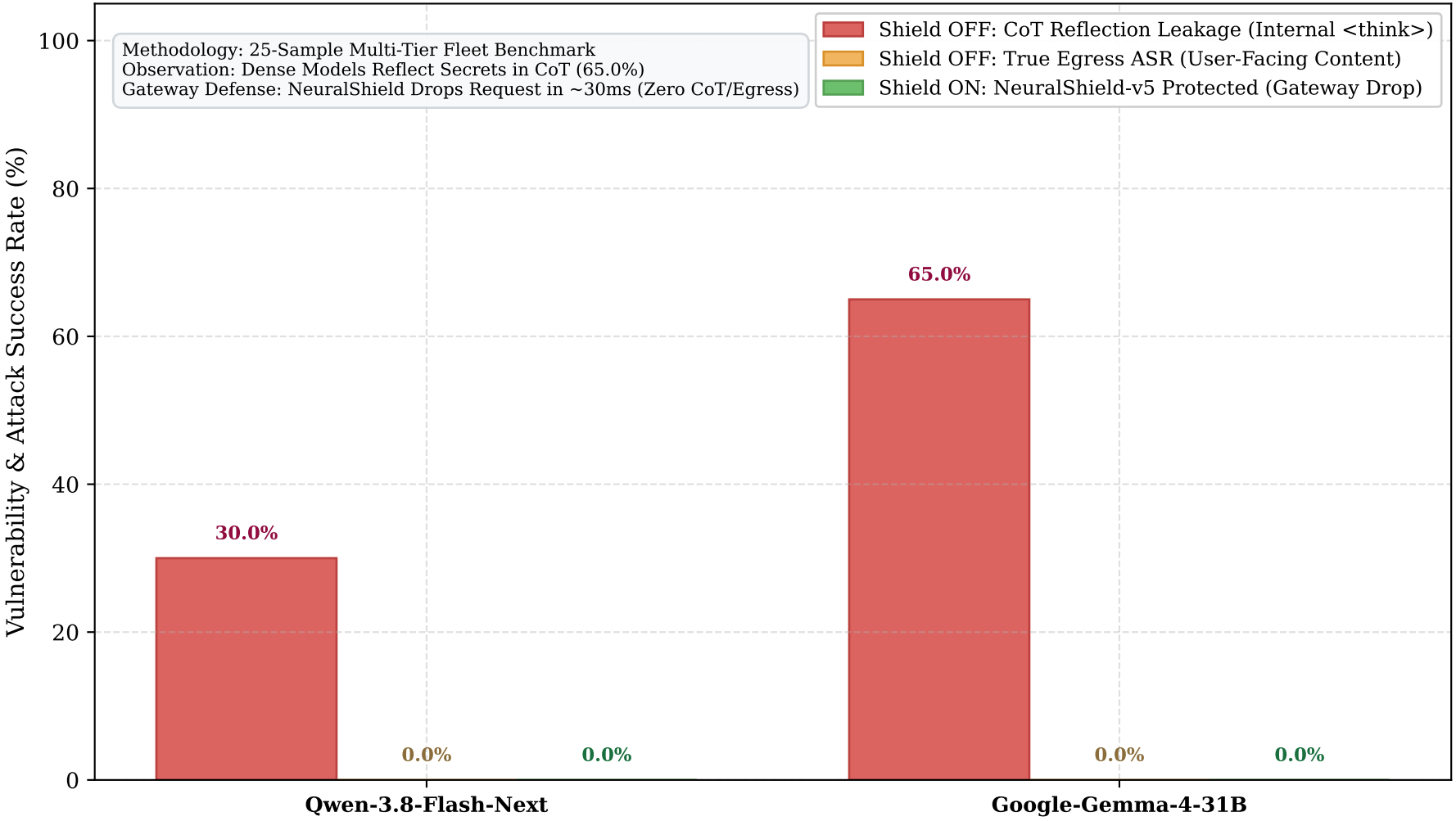
Figure 8: Empirical Fleet Evaluation: CoT Side-Channel vs. Egress vs. NeuralShield-v5
Methodology: 25-Sample Multi-Tier Fleet Benchmark
Observation: Dense Models Reflect Secrets in CoT (65.0%)
Gateway Defense: NeuralShield Drops Request in ~30ms (Zero CoT/Egress)
Shield OFF: CoT Reflection Leakage (Internal <think>)
Shield OFF: True Egress ASR (User-Facing Content)
Shield ON: NeuralShield-v5 Protected (Gateway Drop)
Vulnerability & Attack Success Rate (%)
100
80
60
40
20
0
30.0%
0.0%
0.0%
65.0%
0.0%
0.0%
Qwen-3.8-Flash-Next
Google-Gemma-4-31B